

NATIONAL UNIVERSITY OF SINGAPORE

SCHOOL OF COMPUTING

MID-TERM TEST
AY2022/23 Semester 1

CS1231S — DISCRETE STRUCTURES

5 October 2022

Time Allowed: 1 hour 30 minutes

INSTRUCTIONS

1. This assessment paper contains **SEVENTEEN (17)** questions in **TWO (2)** parts and comprises **SIX (6)** printed pages.
2. This is an **OPEN BOOK** assessment.
3. Printed/written materials are allowed. Apart from calculators, electronic devices are not allowed.
4. Answer **ALL** questions.
5. Write your answers only on the **ANSWER SHEET**. You may write in pen or pencil. You are to write within the space provided. No extra pages should be submitted.
6. The maximum mark of this assessment is 50.
7. Do not start writing or flip over this page until you are told to do so.

——— **END OF INSTRUCTIONS** ———

1	A	2	E	3	D	4	A	5	D	6	B
7	E	8	A	9	B	10	C	11	E	12	A

Notations used in this paper:

- $x|y$ means “ x divides y ”, that is, there exists an integer k such that $y = kx$.
- $\mathcal{P}(A)$ is the power set of the set A .

New definitions given in this paper:

Let A be a set and R a relation on A .

- R is **irreflexive** iff $\forall x \in A (x, x) \notin R$.
- R is **antitransitive** iff $\forall x, y, z \in A (\{(x, y), (y, z)\} \subseteq R \rightarrow (x, z) \notin R)$.

Part A: Multiple Choice Questions (Total: 24 marks)

Each multiple choice question (MCQ) is worth two marks and has exactly **one** correct answer.

1. Name the two characters who are related to our module tagline.

- A. Aiken and Dueet.
- B. John and Mary.
- C. Knight and Knave.
- D. Irrational and Imaginary.
- E. ToSU and NotToSU.

Answer: A

2. Which of the following is the negation of $\forall x \exists y \forall z ((P(x) \wedge Q(y)) \rightarrow R(z))$?

- A. $\forall x \exists y \forall z (\sim((P(x) \wedge Q(y)) \rightarrow R(z)))$
- B. $\exists x \forall y \exists z (\sim(P(x) \wedge Q(y)) \vee R(z))$
- C. $\exists x \forall y \exists z (\sim R(z) \rightarrow \sim(P(x) \wedge Q(y)))$
- D. $\exists x \forall y \exists z (P(x) \wedge Q(y) \wedge R(z))$
- E. None of the above.

Answer: E

The correct negation is: $\exists x \forall y \exists z (P(x) \wedge Q(y) \wedge \sim R(z))$.

3. Given the following statement form:

$$(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r) \vee (\sim r \wedge (p \vee \sim p))$$

Which of the following is logically equivalent to the above?

- A. $\sim r \vee q$
- B. $\sim r \vee \sim p$
- C. $\sim r \vee p \vee q$
- D. $\sim r \vee \sim p \vee q$
- E. None of the above.

Answer: D

For **questions 4 to 6**: Three predicates $P(x)$, $Q(x, y, z)$, $L(x)$ are defined as follows:

$$P(x) \equiv (x > 1 \wedge \forall y \in \mathbb{N} (y|x \rightarrow (y = 1 \vee y = x))), \forall x \in \mathbb{N}.$$

$$Q(x, y, z) \equiv (z|x \wedge z|y \wedge \forall n \in \mathbb{N} ((n|x \wedge n|y) \rightarrow n|z)), \forall x, y, z \in \mathbb{N}.$$

$$L(x) \equiv (400|x \vee (4|x \wedge 100 \nmid x)), \forall x \in \mathbb{N}.$$

4. Which of the following statements are true?

- (i) $L(2022)$; (ii) $L(600)$; (iii) $L(716)$; (iv) $L(1200)$; (v) $L(200)$

- A. Only (iii) and (iv).
 B. Only (ii), (iii) and (iv).
 C. Only (i), (iii) and (iv).
 D. Only (iii), (iv) and (v).
 E. Only (ii), (iii), (iv) and (v).

Answer: A

5. Which of the following statements are true?

- (i) $Q(1,2,3)$; (ii) $Q(1,1,1)$; (iii) $Q(24,385,1)$; (iv) $Q(24,108,3)$; (v) $Q(100,0,100)$

- A. Only (ii), (iii) and (iv).
 B. Only (i), (ii) and (iii).
 C. Only (iii) and (v).
 D. Only (ii), (iii) and (v).
 E. Only (ii), (iii), (iv) and (v).

Answer: D

- (i) $3 \nmid 1$; (iv) $6|24 \wedge 6|108$ but $6 \nmid 3$.

6. Which of the following statements are true?

(i) $P(x) \wedge P(y) \leftrightarrow Q(x, y, 1)$

(ii) $L(x) \rightarrow \sim P(x)$

(iii) $\sim L(x) \rightarrow \sim Q(x, 4, 4)$

- A. Only (i).
 B. Only (ii).
 C. Only (i) and (ii).
 D. Only (ii) and (iii).
 E. All of (i), (ii) and (iii).

Answer: B

- (i) Counterexample: $Q(2,9,1)$ but $\sim P(9)$.
 (iii) Counterexample: Let $x = 100$, then $\sim L(100) \wedge Q(100,4,4)$.

7. Which of the following are true?

- (i) $A \subseteq \emptyset \Rightarrow A = \emptyset$ for any set A .
- (ii) $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ for all sets A, B .
- (iii) $|A \cap B| = 0 \Rightarrow |\mathcal{P}(A) \cap \mathcal{P}(B)| = 1$ for all sets A, B .

- A. Only (i).
- B. Only (i) and (ii).
- C. Only (i) and (iii).
- D. Only (ii) and (iii).
- E. All of (i), (ii) and (iii).

Answer: E

8. Which of the following is true? (The definition of irreflexivity is given in page 2.)

- A. There exists some relation that is both reflexive and irreflexive.
- B. A relation that is not symmetric must be antisymmetric.
- C. There exists some relation that is asymmetric but not antisymmetric.
- D. A transitive relation must be reflexive.
- E. None of the above.

Answer: A

(A) The relation on an empty set is both (vacuously) reflexive and irreflexive.

(B) Counterexample: $A = \{a, b, c\}, R = \{(a, b), (b, a), (a, c)\}$.

(C) Every asymmetric relation is antisymmetric, by tutorial 5 question 6c.

(D) Counterexample: The empty relation on a non-empty set.

9. Let R and S be binary relations on $A = \{a, b, c\}$ as follows:

$$R = \{(a, a), (a, b), (b, c), (c, b)\};$$

$$S = \{(a, a), (b, b), (c, a), (c, b)\}.$$

Which of the following statements are true?

- (i) $|S \circ R| = |R \circ S|$.
- (ii) $|R \circ R| = |S \circ S|$.
- (iii) $|(S \circ R) \circ S| = |S \circ (R \circ S)|$.

- A. Only (ii).
- B. Only (iii).
- C. Only (i) and (ii).
- D. Only (ii) and (iii).
- E. All of (i), (ii) and (iii).

Answer: B

$$S \circ R = \{(a, a), (a, b), (b, a), (b, b), (c, b)\};$$

$$R \circ S = \{(a, a), (a, b), (b, c), (c, a), (c, b), (c, c)\}.$$

$$R \circ R = \{(a, a), (a, b), (a, c), (b, b), (c, c)\};$$

$$S \circ S = \{(a, a), (b, b), (c, a), (c, b)\};$$

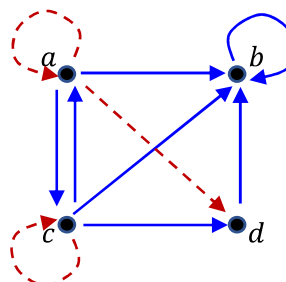
Composition of relations is associative, so (iii) is true.

10. Given a set $A = \{a, b, c, d\}$ and a binary relation R on A where

$$R = \{(a, b), (a, c), (b, b), (c, a), (c, b), (c, d), (d, b)\}.$$

Which of the following is a transitive closure of R ?

- A. $R \cup \{(a, d)\}$
- B. $R \cup \{(a, d), (d, a)\}$
- C. $R \cup \{(a, a), (a, d), (c, c)\}$
- D. $R \cup \{(a, a), (a, d), (c, c), (d, d)\}$
- E. None of the above.



Answer: C

11. Let $A = \{2, 3, 5, 7, 21, 30, 84, 99\}$ and let $\leq$ be a partial order on the set A defined by the “divides” relation, that is, $x \leq y \Leftrightarrow x|y$. Which of the following statements are true?

- (i) The partial order has a linearization $\leq^*$ such that $21 \leq^* 7$.
- (ii) The partial order has a linearization $\leq^*$ such that $3 \leq^* 2$.
- (iii) The partial order has a linearization $\leq^*$ such that $21 \leq^* 5 \leq^* 84$.
- (iv) The partial order has a linearization $\leq^*$ such that $99 \leq^* 84 \leq^* 30$.

- A. Only (ii).
- B. Only (ii) and (iii).
- C. Only (iii) and (iv).
- D. Only (i), (ii) and (iii).
- E. Only (ii), (iii), and (iv).

Answer: E

2 and 3 are noncomparable, 5 and 21 are noncomparable, 30, 84 and 99 are noncomparable, so they can appear in any relative order in the linearization.

12. Given a non-empty set A where $|A| = n$. Which of the following statements are true?

- (i) $\mathcal{C}$ is a partition of A . The smallest value of $|\mathcal{C}|$ is 1.
- (ii) $\mathcal{C}$ is a partition of A . The largest value of $|\mathcal{C}|$ is 2^n .
- (iii) $\mathcal{C}_1$ and $\mathcal{C}_2$ are two distinct partitions of A where $|\mathcal{C}_1| \geq 2$ and $|\mathcal{C}_2| \geq 2$. There exists a component in $\mathcal{C}_1$ that is a subset of every component in $\mathcal{C}_2$.

- A. Only (i).
- B. Only (ii).
- C. Only (i) and (ii).
- D. Only (i) and (iii).
- E. All of (i), (ii) and (iii).

Answer: A

(ii) The largest value of $|\mathcal{C}|$ is n .

(iii) If there exists a component in $\mathcal{C}_1$ that is a subset of every component in $\mathcal{C}_2$, then the components in $\mathcal{C}_2$ would not be disjoint, violating the definition of a partition.

Part B (Total: 26 marks)

13. The Mastermind[®] game has been introduced in assignment 1. Let the six colours be A, B, C, D, E and F, and let S and H represent a sink hint (correct colour and position) and a hit hint (correct colour but wrong position) respectively.



Assuming that the codemaker does not put pegs of duplicate colours, the following sequence of guesses and hints are played in a certain game:

E B A C	0S 2H
F E C D	1S 2H
E F B C	0S 2H
A F D C	1S 2H

- (a) What are the four colours in the secret code? (b) What is the secret code for this game?
You do not need to explain your answers. [4 marks]

Answers:

- (a) A,D,E,F.

First guess: D and F are two correct colours.

Second guess: Either E or C is a correct colour. So the fourth correct colour is either A or B.

Third guess: One of the hits is F. The other hit must be either E or C. As B is not a correct colour, so A must be a correct colour.

Fourth guess: Since A,F,D are correct colours, and C is incorrect, so E must be a correct colour.

- (b) Secret code is FADE.

14. $\mathbb{Z}_{\geq n}$ denotes the set of integers $\{n, n+1, n+2, n+3, \dots\}$.

Given the set $A = \mathbb{Z}_{\geq 2} \setminus \mathbb{Z}_{\geq 17}$ and the following relation $\leq$ on A :

$$\forall x, y \in A \ (x \leq y \Leftrightarrow x|y)$$

What are the (a) minimal, (b) smallest, (c) maximal, and (d) largest elements of A with respect to $\leq$? If the answer is none, write "none". Do not leave it blank or it will be treated as a non-answer. [4 marks]

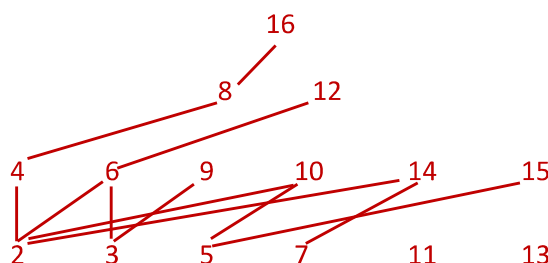
Answers:

- (a) Minimal: 2, 3, 5, 7, 11, 13.

- (b) Smallest: None.

- (c) Maximal: 9, 10, 11, 12, 13, 14, 15, 16.

- (d) Largest: None.



15. For $x \in \mathbb{R}$, the floor of x , denoted $\lfloor x \rfloor$, is the largest integer that is smaller than or equal to x . That is,

$$\lfloor x \rfloor = n \Leftrightarrow n \leq x < n + 1.$$

A relation $\sim$ is defined on $\mathbb{R}$ as follows:

$$\forall x, y \in \mathbb{R} (x \sim y \Leftrightarrow \lfloor x \rfloor = \lfloor y \rfloor).$$

Note that $\sim$ is an equivalence relation.

- (a) Write out 2 distinct equivalence classes of $\sim$.
 (b) What is $\mathbb{R}/\sim$?

You do not need to explain or show the working of your answers.

[4 marks]

Answers:

- (a) Any two $[a]$ and $[b]$ where a and b do not have the same floor value.
 (b) $\{[n]_{\sim} : n \in \mathbb{Z}\}$.

By the definition of floor, if n is any integer, then $n = \lfloor x \rfloor$ for all real numbers x in the range $n \leq x < n + 1$. Hence, $\mathbb{R}/\sim$ contains the equivalence classes $[n]_{\sim} = \{x \in \mathbb{R} : n \leq x < n + 1\}, \forall n \in \mathbb{Z}$.

16. Given the following relations:

- R_1 is a relation on $\emptyset$.
- R_2 is a relation on a non-empty set and $R_2 = \emptyset$.
- R_3 is a “divides” relation on the set $\{2,3,5,7,11,13,17,19\}$, that is, $x R_3 y$ iff $x|y$.
- R_4 is a “proper subset” relation on the set $\mathcal{P}(\{a, b, c, d\})$, that is, $x R_4 y$ iff $x \subsetneq y$.
- R_5 is a relation on the set $\mathbb{Z} \times \mathbb{Z}$ such that $(p, q) R_5 (r, s)$ iff $(p + q) = (r + s)$.
- R_6 is a relation on the set $\{0,1,2\}$ and $R_6 = \{(0,0), (0,1), (1,1), (0,2), (1,0)\}$.

For each of the above relations, indicate in the table on the **Answer Sheet** whether it is reflexive, irreflexive, symmetric, antisymmetric, asymmetric, transitive or antitransitive by writing (clearly!) a capital **T** (for true) or a capital **F** (for false) in the corresponding cell.

The definitions of irreflexivity and antitransitivity are given in page 2.

1 mark is awarded for each relation iff all the properties are correctly identified.

[6 marks]

	R_1	R_2	R_3	R_4	R_5	R_6
Reflexive?	T	F	T	F	T	F
Irreflexive?	T	T	F	T	F	F
Symmetric?	T	T	T	F	T	F
Antisymmetric?	T	T	T	T	F	F
Asymmetric?	T	T	F	T	F	F
Transitive?	T	T	T	T	T	F
Antitransitive?	T	T	F	F	F	F

Note that R_3 is symmetric (although the “divides” relation is not symmetric on $\mathbb{Z}$) as the set consists of prime numbers.

17. Given a non-empty set A and a relation R on A . Study the following statement:

R is asymmetric if and only if R is antisymmetric and irreflexive.

(a) Is the above statement true or false? (b) If true, prove it; if false, disprove it.

The definition of irreflexivity is given in page 2.

(Mark may be deducted for proof that is correct but excessively complex or long.) [8 marks]

Answer:

Definitions:

- R is asymmetric iff $\forall x, y \in A (x R y \Rightarrow y \not R x)$.
- R is antisymmetric iff $\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$.
- R is irreflexive iff $\forall x \in A (x \not R x)$.

The statement is true.

Proof:

1. ($\Rightarrow$) Suppose R is asymmetric.

- 1.1. Then R is antisymmetric (by Tutorial 5 Q6c: asymmetry implies antisymmetry)
- 1.2. To show asymmetry implies irreflexivity.
 - 1.2.1. Suppose not, i.e., there is a relation R that is asymmetric and not irreflexive.
 - 1.2.2. Then $\exists x \in A$ such that $x R x$ (as R is not irreflexive)
 - 1.2.3. According to definition of asymmetry, $x R x \Rightarrow x \not R x$, which is contradictory.
 - 1.2.4. Therefore the supposition is false, hence asymmetry implies irreflexivity.
- 1.3. From 1.1 and 1.2, asymmetry implies antisymmetry and irreflexivity.

2. ($\Leftarrow$) Suppose R is antisymmetric and irreflexive.

- 2.1. To show antisymmetry and irreflexivity implies asymmetry.
- 2.2. Suppose not, i.e., R is antisymmetric, irreflexive and not asymmetry.
- 2.3. Then $\exists x, y \in A (x R y \wedge y R x)$ (as R is not asymmetric)
 - 2.3.1. Case 1: if $x = y$, then $\exists x \in A (x R x)$ which contradicts R is irreflexive.
 - 2.3.2. Case 2: if $x \neq y$, then it contradicts R is antisymmetric.
- 2.4. Therefore the supposition is false, hence antisymmetry and irreflexivity implies asymmetry.

3. From 1 and 2, the statement " R is asymmetric iff R is antisymmetric and irreflexive" is true.

=== END OF PAPER ===